DraftBench — Special Education Resource

Curriculum Aligned & Autism Support Framework

Kinematics Basics

Subject: Physics | **Grade:** 11 | **Language:** ENGLISH | **Duration:** 30 mins

CDC Outcome: null

Visual Schedule

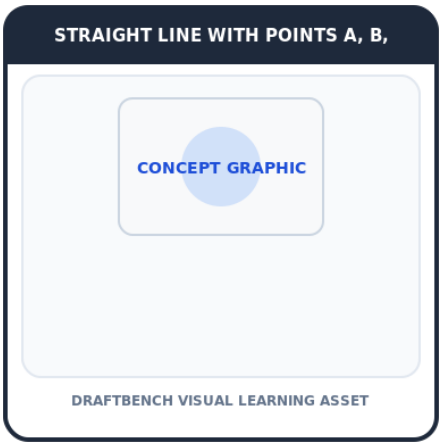
• **First** (5 mins): clock

• **Then** (10 mins): whiteboard

• **Last** (10 mins): handshake

• **Done** (5 mins): checkmark

Lesson Slides



Slide 1: Introduction to Kinematics

• Examine movement

• Without external forces

• Use coordinates

Kinematics studies how objects move. We look at motion without forces acting on them. We use coordinates to describe positions.

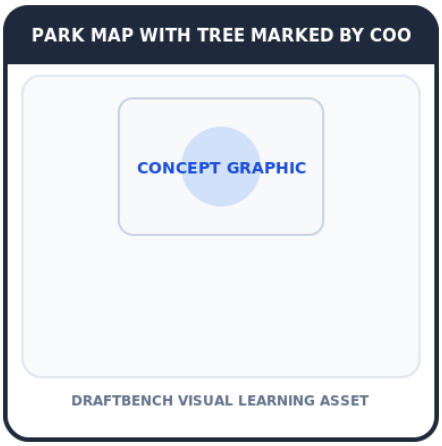
Example: Objects moving on a straight line

Key words: motion: how objects move

Teacher Script: "Today we will learn about motion without forces."

Communication Access: point to coordinates

Sensory Note: null



Slide 2: Position Describes Location

• Position is where

• Use coordinates

• Exact spot

Position tells us where an object is. We use coordinates to find exact spots. Coordinates help us pinpoint locations.

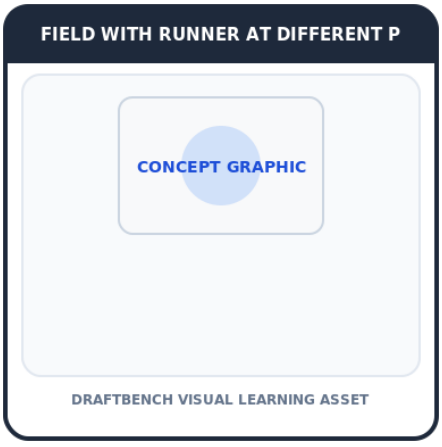
Example: Finding a tree in a park

Key words: coordinates: numbers that show position

Teacher Script: "Position helps us find exact locations. Let's use coordinates."

Communication Access: point to map

Sensory Note: null



Slide 3: Speed Measures How Fast

• Speed is rate

• Use coordinates

• Distance over time

Speed tells us how fast something moves. We use coordinates to measure distance over time. This helps us understand speed.

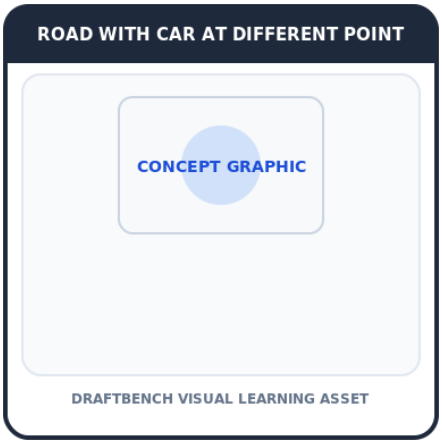
Example: Running around a field

Key words: speed: rate of motion

Teacher Script: "Speed measures how fast we move. Let's use coordinates to track distance."

Communication Access: count steps

Sensory Note: null



Slide 4: Acceleration Measures Change in Speed

• Acceleration is change

• Use coordinates

• Speed over time

Acceleration tells us how speed changes. We use coordinates to measure speed over time. This helps us understand acceleration.

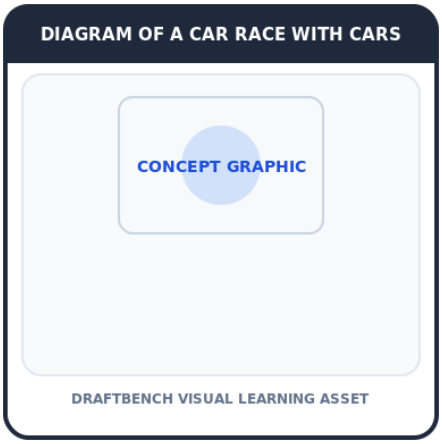
Example: A car speeding up on a road

Key words: acceleration: rate of change in speed

Teacher Script: "Acceleration measures changes in speed. Let's use coordinates to track speed."

Communication Access: point to car

Sensory Note: null



Slide 5: Car Race Example

• Car race

• Use kinematics

• Measure motion

In a car race, we use kinematics to measure motion. This helps us understand how cars move and race.

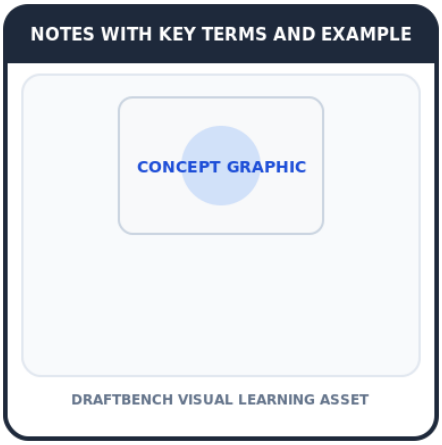
Example: Watching a car race

Key words: kinematics: study of motion

Teacher Script: "Let's use kinematics to understand car races."

Communication Access: point to cars

Sensory Note: null



Slide 6: Review Kinematics Concepts

• Examine motion

• Without forces

• Use coordinates

We reviewed motion without forces. We used coordinates to describe positions. This helps us understand motion better.

Example: Reviewing notes on motion

Key words: motion: how objects move

Teacher Script: "Let's review what we learned about motion."

Communication Access: point to notes

Sensory Note: null

IEP Target

Goal 1

• **English Goal:** Student will develop knowledge related to kinematics at 80% accuracy across 10 parts

• **Nepali Goal:** स्टूडेंट दस पार्ट्यिों में 80% की गतिसे कनिेमेटक्सि की अध्ययन से संबंधति ज्ञान वकिसति करें

• **Baseline Probe:** Student will be given a pre-test with 10 questions on kinematics concepts

- **Measurable Criteria:** 80% accuracy across 10 consecutive parts
- **Progress Monitoring:** Weekly quizzes and observations during class

Accommodations

- Use of visual aids and diagrams

Modifications

- Simplified language and examples

Prompt Hierarchy

independent □ verbal □ visual □ gestural □ physical

Reinforcement Plan

- **Preferred Reinforcer:** Playing a game
- **Schedule:** Token board every 2 correct responses
- **First–Then:** First complete the task, then play a game

Sensory Accommodations

- Use of calming music

Calm-Down Strategy

Take deep breaths and count to 10

Local Context Used

- Use of Nepali examples like car races and fields

Curriculum Sources

No sources retrieved.